AI Playbook and Toolkit for Public Health Departments

ELR, Laboratory Data Quality, and Terminology Mapping

ARC 240

Electronic laboratory reporting supplies critical evidence for infectious disease surveillance, outbreak detection, case investigation, and program monitoring. AI tools that use laboratory data depend on the quality of test codes, result codes, units, specimen information, dates, and mappings to reportable conditions. This module teaches learners how laboratory data quality and terminology mapping affect AI-supported public health work.

Level	applied/foundational
Primary track	Technical Architecture Track
Appears in tracks	technical-architecture

Learning Objectives

- Describe how ELR supports public health surveillance and response.
- Explain why LOINC, SNOMED CT, UCUM, local codes, and reportable condition mappings matter for AI.
- Identify common laboratory data quality problems that can affect AI outputs.
- Assess whether lab data are fit for a specific AI-supported use case.
- Produce a lab data quality checklist for an AI-supported surveillance workflow.

Module Overview

Electronic laboratory reporting supplies critical evidence for infectious disease surveillance, outbreak detection, case investigation, and program monitoring. AI tools that use laboratory data depend on the quality of test codes, result codes, units, specimen information, dates, and mappings to reportable conditions. This module teaches learners how laboratory data quality and terminology mapping affect AI-supported public health work.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Laboratory data are often treated as objective because they come from tests and instruments, but the data that reach public health systems are still mediated by coding, mapping, transmission, and interpretation. A laboratory result can be difficult to use if the test code is local, the result is free text, the units are missing, the specimen source is ambiguous, or the date fields are inconsistent. AI tools can magnify these problems by turning unclear data into confident summaries or classifications.

Terminology mapping is central to reliable public health interpretation. LOINC may identify the test performed, SNOMED CT may represent organisms or coded findings, UCUM may represent units, and local codes may need to be mapped before automated logic can interpret them. Reportable condition mapping connects laboratory evidence to public health action. AI tools should not bypass this informatics work.

For epidemiologists, lab data quality affects case definitions, trend analysis, outbreak signals, and priority setting. For technologists, it affects interface validation, parsing, mapping, and monitoring. For informaticians, it connects laboratory operations, standards, surveillance workflows, and public health authority.

Core Concepts

ELR. Electronic Laboratory Reporting is the electronic transmission of laboratory reports to public health agencies. ELR can improve timeliness and completeness, but it requires interface onboarding, validation, monitoring, and data quality management.

LOINC. LOINC is commonly used to identify laboratory tests and observations. Correct LOINC mapping helps public health systems understand what was tested, not only what result was reported.

SNOMED CT. SNOMED CT is commonly used to represent clinical findings, organisms, and coded results. It can help standardize the interpretation of laboratory evidence and support reportable condition logic.

UCUM. UCUM is a standard for units of measure. Unit consistency is essential when interpreting quantitative results, thresholds, reference ranges, or trends.

Reportable condition mapping. Reportable condition mapping links laboratory tests and results to conditions that may require reporting or investigation. Mapping quality directly affects automated surveillance and AI-supported triage.

Public Health Example

An AI tool that summarizes incoming lab reports for disease investigators may need to identify the test performed, result, organism, specimen source, collection date, result date, performing lab, and ordering provider. If a local test code is mapped incorrectly, the summary may imply evidence for the wrong condition. If units are missing, a quantitative result may be uninterpretable. If the specimen source is omitted, the epidemiologic meaning of the result may change.

A second example is anomaly detection using laboratory trends. If a laboratory changes its coding system or starts reporting a new test type, a model may interpret the change as a disease signal. Without terminology monitoring, staff may investigate an artifact rather than a true public health event.

Technical Deep Dive

Laboratory data quality begins with required fields. At minimum, many workflows need patient identifiers, specimen collection date, test ordered, result, result date, performing lab, provider, specimen source, units when applicable, and reportable condition mapping. The exact requirements depend on the use case. An AI tool used for summarization may need different fields than a tool used for trend detection or case classification.

Terminology mapping should be reviewed as an ongoing process. Local codes may change, new tests may be introduced, and laboratories may report results differently across facilities. Mapping decisions should be versioned, tested, and monitored. When an AI tool depends on mapped data, the mapping version becomes part of the model context and should be documented.

Lab data quality review should include missingness, duplication, test-result pairing, abnormal flags, units, reference ranges, specimen source, date logic, and reportable condition mapping. The review should also identify which issues require human review. Some lab results may be appropriate for automated queueing, while others require epidemiologist interpretation before an AI-generated summary or classification is used.

Risks, Failure Modes, and Guardrails

Misleading standardization. A standard code may be present but mapped incorrectly. Guardrails include mapping review, test cases, and terminology stewardship.

Missing units or specimen details. Incomplete fields can change interpretation. Guardrails include required-field checks and uncertainty flags in AI outputs.

Duplicate or corrected reports. AI summaries may treat corrections as new events. Guardrails include duplicate logic, correction handling, and source display.

Reporting pattern changes. Lab reporting changes may look like disease changes. Guardrails include volume monitoring, partner change logs, and anomaly review.

Application to AI-Supported Workflows

Generative AI may draft lab summaries, but it should preserve source values and avoid interpreting uncertain or missing fields as facts. Predictive analytics may use lab trends, but model monitoring must distinguish reporting artifacts from disease patterns. NLP may interpret free-text results, but extraction accuracy must be validated. Agentic workflows that trigger tasks from lab results require clear thresholds and human escalation for ambiguous results.

Reflection Questions

Which laboratory data quality issues most often affect your surveillance workflows?

Which local codes require mapping before data can be used reliably?

Are test-result pairs consistently represented?

Which missing fields would make AI summarization unsafe?

How would staff detect a reporting change that affects AI outputs?

Practical Exercise

Create a lab data quality checklist for one AI-supported surveillance use case. Include required lab fields, terminology standards, mapping requirements, missingness checks, unit checks, test-result pairing checks, duplicate report review, correction handling, human review points, and escalation criteria.

Expected Artifact or Evidence

Lab data quality checklist

Terminology mapping questions

Required-field list

Data quality issue log

Expected Assignment

- Lab data quality checklist
- Terminology mapping questions
- Required-field list
- Data quality issue log

Knowledge Check

- 1. Why is terminology mapping important for AI-supported surveillance?
- 2. What is LOINC commonly used for?
- 3. Which issue can make a quantitative lab result difficult to interpret?
- 4. What should happen before AI is used to classify lab reports?
- 5. What is strong completion evidence for this module?

References and Resources

- CDC PHIN VADS
- Reportable Condition Mapping Table resources
- APHL laboratory informatics resources
- Jurisdictional ELR onboarding and data quality documentation